

Dimension Reduction for Theta Series of Ideal Lattices

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Abstract. Theta series are fundamental functions that record lattice-vector lengths and connect lattice geometry with Gaussian quantities such as Gaussian mass and the smoothing parameter. For ideal lattices under the canonical embedding, arithmetic coordinates are generally coupled, making fixed-coset theta-series evaluation and related Gaussian computations difficult in high dimensions. We show that an explicit, verifiable relative-basis condition yields an exact factorization of each fixed-coset theta series into lower-dimensional factors.

For each fixed coset, when $n = \ell n_0$, its n -dimensional theta-series computation is replaced by ℓ independent computations in dimension n_0 . Equivalently, the canonical Gram matrix is block diagonal in the relative coordinates, and the factorization holds for every Gaussian parameter without approximation or a smoothing assumption. We verify the condition for several cyclotomic and non-cyclotomic extensions and derive explicit formulas for representative cyclotomic families. The same factorization gives lower-dimensional identities for Gaussian masses, independent Gaussian sampling within a fixed coset, and smoothing-parameter equations. It reduces the dimension of each coset calculation, while the outer sum over the residue classes in I_q remains unchanged. Our implementation achieves speedups of up to $335\times$, $215.45\times$, and $7.9\times$ for fixed-coset theta-series evaluation, numerical smoothing-parameter evaluation, and sequential Gaussian sampling, respectively.

Keywords: Theta Series · Ideal Lattices · Cyclotomic Fields · Lattice Gaussian Sampling · Smoothing Parameter

1 Introduction

Lattices underlie many constructions in post-quantum cryptography, including schemes based on Ring-LWE and ideal lattices [12]. The theta series of a lattice

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is a basic object at the interface between lattice geometry and Gaussian analysis. For a full-rank lattice $\Lambda \subseteq \mathbb{R}^n$, the theta series is defined by

$$\Theta_\Lambda(\tau) := \sum_{\mathbf{x} \in \Lambda} \exp(\pi i \tau \|\mathbf{x}\|^2), \quad \text{Im}(\tau) > 0.$$

The full theta series records the multiplicity of lattice vectors at each squared length. For $s > 0$, its value at i/s^2 is the Gaussian mass $\rho_s(\Lambda)$. On the dual lattice Λ^* , the corresponding Gaussian masses determine the smoothing parameter $\eta_\varepsilon(\Lambda)$ [15,6].

Theta series enter several applications. Gaussian sampling supports lattice trapdoors and digital signatures [11], while smoothing estimates guide the analysis of samplers and their parameter choices [8]. Theta series also appear in the analysis of lattice Gaussian codes [5]. Direct evaluation becomes difficult in high dimensions. Thus a formula that decomposes a theta series into lower-dimensional factors can be useful in all of these applications.

To obtain such a decomposition, we exploit the arithmetic structure of the ideal lattice Λ_I . A number field K with a discrete subring R and an ideal $I \subseteq R$ yields, under the canonical embedding σ_K , the Euclidean lattice $\Lambda_I := \sigma_K(I)$. A subfield $K_0 \subseteq K$, a subring $R_0 \subseteq R$, and a relative basis $b_0, \dots, b_{\ell-1}$ give coordinate blocks, so each $x \in R$ can be written as $x = \sum_j a_j b_j$. For $qR \subseteq I$, each element of I lies in one of the residue classes $c + qR$. We ask whether the canonical squared norm splits as $\|\sigma_K(x)\|^2 = \sum_j \lambda_j \|\sigma_0(a_j)\|^2$ for suitable λ_j . If so, the fixed-coset theta series $\Theta_{c+qR}^K(\tau)$ factors into lower-dimensional theta series. The theta series $\Theta_I(\tau)$ of I is then the sum of these products over $c \in I/qR$, thereby replacing each high-dimensional coset computation with lower-dimensional ones and making otherwise infeasible instances tractable.

Using explicit relative bases, we instantiate this construction for selected cyclotomic and non-cyclotomic families. From the resulting factorizations, we derive corresponding identities for Gaussian mass, coset sampling, and smoothing.

1.1 Our Contributions

Our main contribution is an exact, structure-aware dimension reduction for theta-series computations on q -ary ideal lattices. The reduction is conditional on an explicit norm-separating relative basis; we do not claim that every ideal lattice admits such a basis. More specifically, our contributions are as follows:

(i) General theta-series factorization. We establish a general structural result for theta series on q -ary ideal lattices in Theorem 2. Our field-extension formulation analyzes the canonical norm directly, complementing coordinate and Lee-weight descriptions for general q -ary lattices [5,4]. Its relative-coordinate viewpoint relates to field switching [10].

(ii) Explicit number-field constructions. We give explicit relative bases for cyclotomic extensions, rational radical extensions, and maximal real subfields. The radical cases cover arbitrary relative degrees, and the maximal-real cases

include fields that are not themselves full cyclotomic fields. For cyclotomic conductors, we give power-of-two, odd-prime, and odd-prime-power formulas with one-dimensional, prime-cyclotomic, and $A_{p-1}^\vee$ block forms, consistent with [8].

(iii) **Gaussian and smoothing consequences.** At $\tau = i/s^2$, the factorization gives Gaussian-mass products and independent samples within each fixed coset, and it yields a lower-dimensional smoothing equation through the dual lattice. The outer coset sum remains explicit. In the cyclotomic cases overlapping with [8], we recover the same block decomposition and claim no additional speedup. Our distinction is the theta-series formulation, the explicit criterion, and the extension to additional field families.

(iv) **Implementation and experiments.** We implement full-dimensional and factored fixed-coset evaluations and compare direct and reduced Gaussian sampling with the same SageMath backend [11,18]. Experiments isolate basis change from product decomposition, vary the block dimension, and measure parallelism, showing that the observed gain comes from the product decomposition. Theta-series evaluations use Fincke–Pohst-style enumeration [9].

The factorization reduces fixed-coset computations but does not remove the outer sum over residue classes in I/qR . Our implementation and timing results therefore concern fixed cosets and sampling within a chosen coset, not general coset selection by Gaussian mass or polynomial-time sampling for every ideal lattice.

1.2 Technical Overview

Let $K_0 \subseteq K$ be number fields of degrees n_0 and $n = \ell n_0$ over $\mathbb{Q}$, with orders $R_0 \subseteq R$ and canonical embeddings σ_0, σ_K . For an integer $q \geq 2$, consider an ideal I satisfying $qR \subseteq I \subseteq R$. We first evaluate a fixed coset $c + qR$, where $c \in I$. For T equal to this coset or to I , write

$$\Theta_T^K(\tau) := \sum_{x \in T} \exp(\pi i \tau \|\sigma_K(x)\|^2), \quad \text{Im}(\tau) > 0.$$

The analogous notation over K_0 uses σ_0 .

A relative basis $(b_0, \dots, b_{\ell-1})$ gives $R = \bigoplus_{j=0}^{\ell-1} R_0 b_j$. It preserves the integral coordinates but need not separate the canonical norm. We require positive constants λ_j such that

$$\left\| \sigma_K \left(\sum_{j=0}^{\ell-1} a_j b_j \right) \right\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(a_j)\|^2, \quad a_j \in K_0.$$

This is the norm-separating condition: the relative components are orthogonal, and each carries a scaled copy of the geometry of K_0 . Write $c = \sum_j c_j b_j$, with $c_j \in R_0$. Every element of $c + qR$ is uniquely $\sum_j (c_j + qz_j) b_j$, where the $z_j \in R_0$ vary independently. The norm identity separates the exponential weight, so

absolute convergence gives

$$\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{\ell-1} \Theta_{c_j+qR_0}^{K_0}(\lambda_j\tau) \quad (\text{Theorem 2}).$$

Thus one computation in dimension n becomes ℓ computations in dimension n_0 . The theta series of I is still the sum of these products over I/qR . The factorization reduces the dimension of each summand, without removing the cost of that outer sum.

At $\tau = i/s^2$, the same product gives the Gaussian mass. After normalization, it yields independent samples within a fixed coset, with widths $s/\sqrt{\lambda_j}$ in the smaller lattices. Sampling from the whole ideal additionally requires selecting a coset according to its mass. Poisson summation relates the theta series of the ideal to that of its dual, allowing the same factorization to enter the smoothing-parameter equation in Section 6.

1.3 Related Work

Our work is related to structured ideal lattices, coset decompositions for q -ary lattices, and discrete Gaussian sampling and smoothing.

Ideal lattices, cyclotomic rings, and field switching. Ideal lattices and cyclotomic rings are standard in Ring-LWE cryptography [12]. Gentry, Halevi, Peikert, and Smart introduced relative bases for field switching [10], while prior work developed efficient orthogonalization and sampling methods for structured lattices [13,7]. Espitau, Wallet, and Yu studied cyclotomic sampling and related smoothing estimates [8]. Together, these works establish the structured representations and samplers relevant here. We use relative bases to derive an exact theta-series factorization for fixed cosets under an explicit norm-separating condition, including cases beyond the overlapping cyclotomic constructions.

Coset decompositions and q -ary lattices. Campello and Belfiore used coset decompositions in Gaussian sampling for lattice codes [5]. Bollauf, Lie, and Ling related Gaussian sampling on q -ary lattices to Lee-weight profiles and symmetrized weight enumerators [4]. Our power-of-two formula gives the corresponding weight-enumerator form, but its factorization comes from a relative basis and orthogonality of the canonical norm. This also allows the same argument to cover the ideal-lattice families considered below.

Discrete Gaussian sampling and smoothing. Micciancio and Regev introduced the smoothing parameter in lattice cryptography [15]. Gaussian sampling was developed for lattice trapdoors and related constructions [11], followed by work on parallel, integer, and structured-lattice samplers [17,1,16,13,7]. The complexity of approximating smoothing parameters for general lattices was studied by Chung, Dadush, Liu, and Peikert [6]. These works provide the sampling and smoothing background. We use the same quantities to state exact fixed-coset identities for ideal lattices, while retaining the cost of selecting a residue class.

Our contribution connects these ingredients through an exact fixed-coset theta-series factorization. The outer sum over residue classes remains a separate computational problem.

1.4 Paper Organization

Section 2 introduces the notation, embeddings, lattices, theta series, and smoothing parameter. Section 3 defines norm separation and proves the general factorization results. Section 4 and Section 5 verify the condition for the field families considered here. Section 6 derives Gaussian-mass, sampling, and smoothing consequences, while Section 7 reports the implementation and experiments. The appendices contain explicit bases, longer proofs, and additional computational checks.

2 Preliminaries

2.1 Notation

We write $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ for the integers, rationals, reals, and complex numbers, respectively. For a positive integer u , let $[u] := \{0, \dots, u-1\}$. The Euler totient function is denoted by φ . For a complex number z , we write $\bar{z}$ for its complex conjugate.

For a matrix $\mathbf{A}$, we write $\mathbf{A}^T$ and $\mathbf{A}^*$ for its transpose and conjugate transpose. The identity matrix of dimension d is denoted by $\mathbf{I}_d$. The notation $\sqcup$ indicates a disjoint union.

2.2 Algebraic Number Theory Background

Let K be a number field of degree $n = [K : \mathbb{Q}]$. We recall the canonical embedding and its associated Euclidean structure, following the standard notation of [14, 12].

Definition 1 (Canonical Embedding). *Let $\Sigma_K = \{\sigma_1, \dots, \sigma_n\}$ be the set of embeddings of K into $\mathbb{C}$. The full canonical embedding is $\sigma_K : K \rightarrow \mathbb{C}^n$, $a \mapsto (\sigma_1(a), \dots, \sigma_n(a))$.*

If $\bar{\sigma}$ denotes the embedding $a \mapsto \overline{\sigma(a)}$, then the image of σ_K lies in the n -dimensional real subspace $H_K := \{(z_\sigma)_{\sigma \in \Sigma_K} \in \mathbb{C}^n : z_{\bar{\sigma}} = \overline{z_\sigma} \text{ for every } \sigma \in \Sigma_K\}$. The space H_K has real dimension n , and we regard $\sigma_K(K)$ as embedded in this Euclidean space throughout the paper.

We equip H_K with the Euclidean inner product inherited from $\mathbb{C}^n$,

$$\langle \mathbf{z}, \mathbf{w} \rangle := \operatorname{Re} \left(\sum_{\sigma \in \Sigma_K} z_\sigma \overline{w_\sigma} \right), \quad \|\mathbf{z}\|^2 := \sum_{\sigma \in \Sigma_K} |z_\sigma|^2.$$

Definition 2 (Canonical Norm and Field Trace). *For $a \in K$, we write $\|a\| := \|\sigma_K(a)\|$. The field trace is defined as $\operatorname{Tr}_{K/\mathbb{Q}}(a) := \sum_{\sigma \in \Sigma_K} \sigma(a)$.*

Proposition 1. *Suppose that K admits a field automorphism $x \mapsto \bar{x}$ satisfying $\sigma(\bar{x}) = \overline{\sigma(x)}$ for every embedding $\sigma : K \hookrightarrow \mathbb{C}$ and every $x \in K$. Then for all $a, b \in K$, $\langle \sigma_K(a), \sigma_K(b) \rangle = \text{Tr}_{K/\mathbb{Q}}(a\bar{b})$.*

Proof. By the compatibility assumption, $\text{Tr}_{K/\mathbb{Q}}(a\bar{b}) = \sum_{\sigma: K \hookrightarrow \mathbb{C}} \sigma(a)\overline{\sigma(b)}$. The sum is real, since its terms corresponding to conjugate embeddings occur in conjugate pairs. Taking its real part therefore gives the canonical inner product.

Definition 3 (Gram Matrix). *Let $R \subseteq K$ be an order of rank n , and let $\mathcal{B} = (\beta_1, \dots, \beta_n)$ be a $\mathbb{Z}$ -basis of R . The Gram matrix of the embedded lattice $\sigma_K(R)$ with respect to $\mathcal{B}$ is $\mathbf{\Gamma}_{\mathcal{B}} := (\langle \sigma_K(\beta_i), \sigma_K(\beta_j) \rangle)_{1 \leq i, j \leq n}$.*

For $\mathbf{z} = (z_1, \dots, z_n)^T \in \mathbb{Z}^n$ and $a = \sum_i z_i \beta_i$, we have $\|\sigma_K(a)\|^2 = \mathbf{z}^T \mathbf{\Gamma}_{\mathcal{B}} \mathbf{z}$, so $\mathbf{\Gamma}_{\mathcal{B}}$ represents the canonical quadratic form in the coordinates determined by $\mathcal{B}$. Under a basis change $\mathcal{B}' = \mathcal{B}\mathbf{U}$, with $\mathbf{U} \in \text{GL}_n(\mathbb{Z})$, the Gram matrix transforms as $\mathbf{\Gamma}_{\mathcal{B}'} = \mathbf{U}^T \mathbf{\Gamma}_{\mathcal{B}} \mathbf{U}$. This basis dependence will be exploited later to obtain a block diagonal Gram matrix.

An element $\alpha \in K$ is an algebraic integer if it is a root of a monic polynomial in $\mathbb{Z}[X]$. The set of algebraic integers in K is its ring of integers $\mathcal{O}_K$, which is a free $\mathbb{Z}$ -module of rank n . An order R in K is a subring $R \subseteq \mathcal{O}_K$ containing 1 that is a free $\mathbb{Z}$ -module of rank n . Throughout this paper, R denotes the ambient order containing the ideals under consideration.

Definition 4 (Field Discriminant). *Let $\mathcal{B} = (\beta_1, \dots, \beta_n)$ be an integral basis of $\mathcal{O}_K$. The discriminant of K is $\Delta_K := \det(\text{Tr}_{K/\mathbb{Q}}(\beta_i \beta_j))_{1 \leq i, j \leq n}$, which is independent of the chosen integral basis.*

2.3 Lattice Background

We now apply the canonical embedding to ideals of the ambient order R .

Definition 5 (Ideal Lattice). *A nonzero ideal $I \subseteq R$ is a free $\mathbb{Z}$ -module of rank n . Its canonical embedding $\sigma_K(I)$ is therefore a full-rank lattice in H_K , called an ideal lattice.*

Definition 6 (q -ary Ideal). *For an integer $q \geq 2$, an ideal $I \subseteq R$ is called q -ary if $qR \subseteq I \subseteq R$. The quotient $I_q := I/qR$ is then a finite ideal of R/qR .*

Definition 7 (Ideal Norm). *Let $I \subseteq \mathcal{O}_K$ be a nonzero ideal. Its norm is defined as $N(I) := [\mathcal{O}_K : I] = |\mathcal{O}_K/I|$.*

For $I \subseteq \mathcal{O}_K$, the full canonical embedding gives the ideal lattice identity $\det(\sigma_K(I)) = N(I)\sqrt{|\Delta_K|}$. For a full-rank lattice $\Lambda \subseteq H_K$ with basis $v_1, \dots, v_n$, we define

$$\det(\Lambda) := \sqrt{\det\left(\left(\langle v_i, v_j \rangle\right)_{1 \leq i, j \leq n}\right)}.$$

If $\mathbf{B}$ is the coordinate matrix of this basis with respect to an orthonormal real basis of H_K , then equivalently $\det(\Lambda) = \sqrt{\det(\mathbf{B}^T \mathbf{B})}$.

2.4 Theta Series and Discrete Gaussian Distributions

We introduce theta series and discrete Gaussian distributions with respect to the Euclidean norm defined above.

Definition 8 (Theta Series). *Let $H \subseteq H_K$ be a discrete set for which the following series converges absolutely. Its theta series is*

$$\Theta_H(\tau) := \sum_{\mathbf{x} \in H} \exp(\pi i \tau \|\mathbf{x}\|^2), \quad \text{Im}(\tau) > 0.$$

For $T \subseteq K$, we abbreviate $\Theta_T(\tau) := \Theta_{\sigma_K(T)}(\tau)$.

For $s > 0$ and $\mathbf{x} \in H_K$, define $\rho_s(\mathbf{x}) := \exp(-\pi \|\mathbf{x}\|^2 / s^2)$. For a discrete set S of finite Gaussian mass, let $\rho_s(S) := \sum_{\mathbf{x} \in S} \rho_s(\mathbf{x})$. Then $\rho_s(S) = \Theta_S(i/s^2)$.

Definition 9 (Discrete Gaussian Distribution). *Let A be a discrete set with finite Gaussian mass. The discrete Gaussian distribution over A with parameter $s > 0$ is*

$$D_{A,s}(\mathbf{x}) := \frac{\rho_s(\mathbf{x})}{\rho_s(A)}, \quad \mathbf{x} \in A.$$

We recall the following identities. If S_1, S_2 lie in orthogonal Euclidean subspaces, then $\Theta_{S_1 \oplus S_2}(\tau) = \Theta_{S_1}(\tau) \Theta_{S_2}(\tau)$, $D_{S_1 \oplus S_2, s}(\mathbf{x}_1 + \mathbf{x}_2) = D_{S_1, s}(\mathbf{x}_1) D_{S_2, s}(\mathbf{x}_2)$, where $\mathbf{x}_i \in S_i$. For $\alpha > 0$, $\Theta_{\alpha S}(\tau) = \Theta_S(\alpha^2 \tau)$ and $D_{\alpha S, s}(\alpha \mathbf{x}) = D_{S, s/\alpha}(\mathbf{x})$, where $\mathbf{x} \in S$.

2.5 Smoothing Parameter

The smoothing parameter is defined through the Gaussian mass of the dual lattice and can equivalently be characterized by its theta series.

Definition 10 (Dual Lattice). *For a full-rank lattice $\Lambda \subseteq H_K$, its dual lattice is defined as $\Lambda^* := \{y \in H_K : \langle \mathbf{x}, y \rangle \in \mathbb{Z} \text{ for every } \mathbf{x} \in \Lambda\}$.*

Definition 11 (Smoothing Parameter). *For a full-rank lattice $\Lambda \subseteq H_K$ and $\varepsilon > 0$, its smoothing parameter is $\eta_\varepsilon(\Lambda) := \inf \{s > 0 : \rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) \leq \varepsilon\}$.*

By the relation between Gaussian mass and theta series, $\eta_\varepsilon(\Lambda)$ is equivalently the smallest $s > 0$ satisfying $\Theta_{\Lambda^*}(is^2) - 1 \leq \varepsilon$.

3 Decomposition of Theta Series

Theta series are additive over coset decompositions and factor when the corresponding components are orthogonal. This principle appears in Construction A lattices through coordinate and weight decompositions [5,4], and in cyclotomic Gaussian-sampling constructions [8]. For ideal lattices, however, algebraic coordinates need not be orthogonal under the canonical norm, so such a factorization is not automatic.

This section gives an explicit condition for q -ary ideal-lattice cosets. Let K be a number field, let R be an order in K , and let $I \subseteq R$ be a q -ary ideal. We define a norm-separating relative basis over a subfield K_0 , derive a relative-trace criterion and the associated block-diagonal Gram matrix, and obtain the resulting theta-series factorization. Under this condition, each fixed coset theta series becomes a product of ℓ theta series over K_0 . At $\tau = i/s^2$, the same identity gives the Gaussian-mass and within-coset sampling formulas used in Section 6, while the outer sum over $I_q = I/qR$ remains unchanged.

Recall that $I_q := I/qR$ is finite and that each class $c \in I_q$ corresponds to a coset $c + qR \subseteq I$. Throughout this section, we fix one representative in I for every class in I_q , and use the same symbol c for the class and its representative.

3.1 Coset Decomposition

Since $qR \subseteq I$, the classes in $I_q = I/qR$ index the cosets of qR contained in I . Hence, we have the coset decomposition of I :

$$I = \bigsqcup_{c \in I_q} (c + qR).$$

Lemma 1 (Decomposition of the Theta Series). *For every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$,*

$$\Theta_I(\tau) = \sum_{c \in I_q} \Theta_{c+qR}(\tau).$$

Proof. Since the cosets $c + qR$, indexed by $c \in I_q$, form a disjoint partition of I , we have $\Theta_I(\tau) = \sum_{c \in I_q} \sum_{x \in c+qR} \exp(\pi i \tau \|\sigma_K(x)\|^2) = \sum_{c \in I_q} \Theta_{c+qR}(\tau)$.

At $\tau = i/s^2$, these theta-series identities give the corresponding Gaussian masses.

Let $\mathcal{B} = (\beta_1, \dots, \beta_n)$ be a $\mathbb{Z}$ -basis of R , and let $\mathbf{\Gamma}_B$ be the corresponding Gram matrix. For a fixed $c \in R$, write $c = \sum_{i=1}^n c_i \beta_i$ and let $\mathbf{c} = (c_1, \dots, c_n)^T$. Every element of the coset $c + qR$ has a unique representation $c + q \sum_{i=1}^n z_i \beta_i$, $\mathbf{z} = (z_1, \dots, z_n)^T \in \mathbb{Z}^n$. Therefore,

$$\left\| \sigma_K \left(c + q \sum_{i=1}^n z_i \beta_i \right) \right\|^2 = (\mathbf{c} + q\mathbf{z})^T \mathbf{\Gamma}_B (\mathbf{c} + q\mathbf{z}).$$

For the fixed coset $c + qR$, define

$$Q_c(\mathbf{z}) := (\mathbf{c} + q\mathbf{z})^T \mathbf{\Gamma}_B (\mathbf{c} + q\mathbf{z}), \quad \mathbf{z} \in \mathbb{Z}^n.$$

Then the theta series of the coset is $\Theta_{c+qR}(\tau) = \sum_{\mathbf{z} \in \mathbb{Z}^n} \exp(\pi i \tau Q_c(\mathbf{z}))$. The choice of basis determines which coordinates are coupled by Q_c . A block-diagonal Gram matrix is sufficient for a product decomposition: cross terms must vanish between distinct blocks, although they may remain within each block. A nonzero off-diagonal entry in an arbitrary basis does not by itself rule out such a decomposition, because the block structure is basis dependent. We therefore look for a basis adapted to a subfield of K .

3.2 Relative Bases

Let $K_0 \subseteq K$ be a subfield, and let $R_0 \subseteq K_0$ be an order satisfying $R_0 \subseteq R$. Write $n_0 = [K_0 : \mathbb{Q}]$, $\ell = [K : K_0]$, $n = \ell n_0$, and let σ_0 denote the full canonical embedding of K_0 . An R_0 -basis of R exists precisely when R is free of rank ℓ as an R_0 -module. This is an assumption on the chosen pair of orders and does not follow from $R_0 \subseteq R$.

Relative bases with controlled geometry under the canonical embedding arise in field switching for cyclotomic homomorphic encryption [10, Sec. 2.1.5]. Here, the same module viewpoint is used for a different computational purpose: we seek coordinates in which the Gaussian weight inside a fixed coset $c + qR$ separates exactly into lower-dimensional factors.

Definition 12 (Relative Basis). *An ordered set $\mathbf{b} = (b_0, \dots, b_{\ell-1}) \subseteq R$ is an R_0 -basis of R if every $a \in R$ has a unique representation $a = \sum_{j=0}^{\ell-1} a_j b_j$ with $a_j \in R_0$. Equivalently, $R = \bigoplus_{j=0}^{\ell-1} R_0 b_j$.*

A relative basis gives unique R_0 -coordinates for every element of R , but this algebraic decomposition does not imply that the corresponding components are orthogonal under the canonical inner product. We isolate the additional norm property needed for dimension reduction.

Definition 13 (Norm-Separating Relative Basis). *Let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be an R_0 -basis of R . We call $\mathbf{b}$ norm-separating if there exist $\lambda_0, \dots, \lambda_{\ell-1} \in \mathbb{R}_{>0}$ such that*

$$\left\| \sigma_K \left(\sum_{j=0}^{\ell-1} a_j b_j \right) \right\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(a_j)\|^2$$

for all $a_0, \dots, a_{\ell-1} \in K_0$. The values $\lambda_0, \dots, \lambda_{\ell-1}$ are called the scaling factors of $\mathbf{b}$.

The following lemma gives a convenient sufficient condition in terms of relative traces. We use the standard trace properties from [14]. An exact finite existence test is given in Appendix D.

Lemma 2 (Trace Criterion for Norm Separation). *Assume that K admits a field automorphism $x \mapsto \bar{x}$ preserving K_0 , with $\bar{\bar{x}} = x$, such that $\sigma(\bar{x}) = \overline{\sigma(x)}$ for every embedding $\sigma : K \hookrightarrow \mathbb{C}$ and every $x \in K$. Let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be an R_0 -basis of R . Suppose that there exist $\lambda_0, \dots, \lambda_{\ell-1} \in \mathbb{Q}_{>0}$ such that*

$$\mathrm{Tr}_{K/K_0}(b_i \bar{b}_j) = \begin{cases} \lambda_i, & i = j, \\ 0, & i \neq j. \end{cases}$$

Then $\mathbf{b}$ is norm-separating with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. Equivalently, for every $a_0, \dots, a_{\ell-1} \in K_0$,

$$\left\| \sigma_K \left(\sum_{j=0}^{\ell-1} a_j b_j \right) \right\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(a_j)\|^2.$$

Proof. Set $x = \sum_{j=0}^{\ell-1} a_j b_j$. Compatibility with every embedding gives

$$\|\sigma_K(x)\|^2 = \sum_{\sigma: K \hookrightarrow \mathbb{C}} \sigma(x) \overline{\sigma(x)} = \text{Tr}_{K/\mathbb{Q}}(x\bar{x}).$$

By transitivity of the field trace and K_0 -linearity of the relative trace,

$$\text{Tr}_{K/K_0}(x\bar{x}) = \sum_{i,j=0}^{\ell-1} a_i \bar{a}_j \text{Tr}_{K/K_0}(b_i \bar{b}_j) = \sum_{j=0}^{\ell-1} \lambda_j a_j \bar{a}_j.$$

Consequently,

$$\|\sigma_K(x)\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \text{Tr}_{K_0/\mathbb{Q}}(a_j \bar{a}_j) = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(a_j)\|^2,$$

where the last equality follows from the same compatibility condition restricted to K_0 .

In Lemma 2, requiring the diagonal trace values to lie in $\mathbb{Q}_{>0}$ ensures that they act as the same scalar under every embedding of K_0 . Nonrational trace values in K_0 would generally produce embedding-dependent weighted norms. Definition 13, however, allows arbitrary positive real scaling factors whenever the norm-separation identity can be verified directly.

Lemma 2 shows that relative-trace orthogonality is sufficient for norm separation. The arguments below use only the norm-separation identity itself, so the same results also apply when that identity is verified directly from the embeddings.

Corollary 1 (Relative Component Orthogonality). *Let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be a norm-separating R_0 -basis of R , with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. Then, for $a, b \in K_0$ and $0 \leq i, j < \ell$,*

$$\langle \sigma_K(ab_i), \sigma_K(bb_j) \rangle = \begin{cases} \lambda_i \langle \sigma_0(a), \sigma_0(b) \rangle, & i = j, \\ 0, & i \neq j. \end{cases}$$

In particular, the real subspaces $V_j := \text{span}_{\mathbb{R}}(\sigma_K(K_0 b_j))$, $0 \leq j < \ell$, are mutually orthogonal.

Proof. Suppose first that $i \neq j$. The norm-separation identity gives

$$\|\sigma_K(ab_i + bb_j)\|^2 = \lambda_i \|\sigma_0(a)\|^2 + \lambda_j \|\sigma_0(b)\|^2.$$

The two terms on the right are respectively $\|\sigma_K(ab_i)\|^2$ and $\|\sigma_K(bb_j)\|^2$. The polarization identity therefore gives $\langle \sigma_K(ab_i), \sigma_K(bb_j) \rangle = 0$. For $i = j$, applying the same identity to $(a+b)b_i$, ab_i , and bb_i gives

$$\langle \sigma_K(ab_i), \sigma_K(bb_i) \rangle = \lambda_i \langle \sigma_0(a), \sigma_0(b) \rangle.$$

Thus, a norm-separating relative basis decomposes the canonical Euclidean structure into ℓ mutually orthogonal scaled copies of the structure of K_0 , which we express in integral coordinates below.

3.3 Block-Diagonal Gram Matrix

Let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be a norm-separating R_0 -basis of R , with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. Corollary 1 shows that different relative components are orthogonal. Let $\mathcal{A} = (\alpha_1, \dots, \alpha_{n_0})$ be a $\mathbb{Z}$ -basis of R_0 , with Gram matrix

$$\mathbf{\Gamma}_{\mathcal{A}} := (\langle \sigma_0(\alpha_u), \sigma_0(\alpha_v) \rangle)_{1 \leq u, v \leq n_0}.$$

Combining $\mathcal{A}$ with $\mathbf{b}$ gives the ordered $\mathbb{Z}$ -basis

$$\mathcal{B} := (\alpha_1 b_0, \dots, \alpha_{n_0} b_0, \alpha_1 b_1, \dots, \alpha_{n_0} b_1, \dots, \alpha_1 b_{\ell-1}, \dots, \alpha_{n_0} b_{\ell-1})$$

of R . The basis vectors are grouped according to $R_0 b_0, \dots, R_0 b_{\ell-1}$. By Corollary 1, different groups are orthogonal and the j -th group is scaled by λ_j , giving the following block-diagonal form.

Theorem 1 (Block-Diagonal Gram Matrix). *Let $\mathbf{b}$ be a norm-separating R_0 -basis of R with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. With $\mathcal{A}$, $\mathcal{B}$, and $\mathbf{\Gamma}_{\mathcal{A}}$ as above, the Gram matrix of $\sigma_K(R)$ with respect to $\mathcal{B}$ is*

$$\mathbf{\Gamma}_{\mathcal{B}} = \begin{pmatrix} \lambda_0 \mathbf{\Gamma}_{\mathcal{A}} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \lambda_{\ell-1} \mathbf{\Gamma}_{\mathcal{A}} \end{pmatrix},$$

with ℓ diagonal blocks of dimension n_0 .

Proof. For $0 \leq i, j < \ell$ and $1 \leq u, v \leq n_0$, consider the entry $\langle \sigma_K(\alpha_u b_i), \sigma_K(\alpha_v b_j) \rangle$ of $\mathbf{\Gamma}_{\mathcal{B}}$. By Corollary 1,

$$\langle \sigma_K(\alpha_u b_i), \sigma_K(\alpha_v b_j) \rangle = \begin{cases} \lambda_i \langle \sigma_0(\alpha_u), \sigma_0(\alpha_v) \rangle, & i = j, \\ 0, & i \neq j. \end{cases}$$

Hence all off-diagonal blocks are zero. For $i = j$, the block indexed by $R_0 b_i$ is exactly $\lambda_i \mathbf{\Gamma}_{\mathcal{A}}$. This therefore proves

$$\mathbf{\Gamma}_{\mathcal{B}} = \begin{pmatrix} \lambda_0 \mathbf{\Gamma}_{\mathcal{A}} & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \lambda_{\ell-1} \mathbf{\Gamma}_{\mathcal{A}} \end{pmatrix}$$

as claimed.

Theorem 1 separates the n -dimensional quadratic form into ℓ blocks of dimension n_0 . The matrix $\mathbf{\Gamma}_{\mathcal{A}}$ need not be diagonal, so cross terms may remain within a block. Only cross terms between distinct relative components vanish. The scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$ need not be equal.

Since $q\mathcal{B}$ is a $\mathbb{Z}$ -basis of qR , $\mathbf{\Gamma}_{q\mathcal{B}} = q^2 \mathbf{\Gamma}_{\mathcal{B}}$, so multiplication by q preserves the relative block structure used below for $c + qR$.

3.4 Factorization of Coset Theta Series

For $T \subseteq K$, write $\Theta_T^K(\tau) := \Theta_{\sigma_K(T)}(\tau)$, and for $T \subseteq K_0$, write $\Theta_T^{K_0}(\tau) := \Theta_{\sigma_0(T)}(\tau)$. Fix $c \in R$. By Definition 12, there are unique $c_0, \dots, c_{\ell-1} \in R_0$ such that $c = \sum_{j=0}^{\ell-1} c_j b_j$. The relative-coordinate map gives the bijective description

$$c + qR = \left\{ \sum_{j=0}^{\ell-1} (c_j + qz_j) b_j : z_0, \dots, z_{\ell-1} \in R_0 \right\}.$$

The variables z_j range independently over R_0 , and the norm-separation identity in Definition 13 separates the corresponding quadratic exponent. This gives the following product formula.

Theorem 2 (Factorization of Coset Theta Series). *Let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be a norm-separating R_0 -basis of R , with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$, and write $c = \sum_{j=0}^{\ell-1} c_j b_j$, $c_j \in R_0$. Then, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$,*

$$\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{\ell-1} \Theta_{c_j+qR_0}^{K_0}(\lambda_j \tau).$$

Proof. Every element of $c + qR$ has a unique representation $y = \sum_{j=0}^{\ell-1} (c_j + qz_j) b_j$, $z_j \in R_0$. By Definition 13, $\|\sigma_K(y)\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(c_j + qz_j)\|^2$. Substituting this identity into the definition of the theta series gives

$$\begin{aligned} \Theta_{c+qR}^K(\tau) &= \sum_{(z_0, \dots, z_{\ell-1}) \in R_0^\ell} \exp \left(\pi i \tau \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(c_j + qz_j)\|^2 \right) \\ &= \sum_{(z_0, \dots, z_{\ell-1}) \in R_0^\ell} \prod_{j=0}^{\ell-1} \exp \left(\pi i \lambda_j \tau \|\sigma_0(c_j + qz_j)\|^2 \right). \end{aligned}$$

Since $\text{Im}(\tau) > 0$ and $\lambda_j > 0$ for every j , the series is absolutely convergent. Therefore, the sums over $z_0, \dots, z_{\ell-1}$ can be separated, and

$$\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{\ell-1} \left(\sum_{z_j \in R_0} \exp \left(\pi i \lambda_j \tau \|\sigma_0(c_j + qz_j)\|^2 \right) \right) = \prod_{j=0}^{\ell-1} \Theta_{c_j+qR_0}^{K_0}(\lambda_j \tau).$$

Theorem 2 reduces one n -dimensional coset theta series to ℓ factors of dimension n_0 . In particular, at $\tau = i/s^2$,

$$\Theta_{c+qR}^K(i/s^2) = \prod_{j=0}^{\ell-1} \Theta_{c_j+qR_0}^{K_0}(i\lambda_j/s^2).$$

The distributional consequence is exact as well. If $X_0, \dots, X_{\ell-1}$ are independent K_0 -valued random variables satisfying $\sigma_0(X_j) \sim D_{\sigma_0(c_j + qR_0), s/\sqrt{\lambda_j}}$, then

$$\sigma_K\left(\sum_{j=0}^{\ell-1} X_j b_j\right) \sim D_{\sigma_K(c + qR), s}.$$

This is the ideal-lattice counterpart of the coordinatewise coset factorization used for Construction A lattices in [5, Propositions 1–2] and [4, Lemma 1 and Theorems 1–2]. In those works, a coset of $q\mathbb{Z}^n$ splits into one-dimensional shifted-integer factors, and the coset weights are organized by Hamming or Lee weight profiles. Here, the blocks are the embedded R_0 -modules $R_0 b_j$, each factor has dimension n_0 , and no weight-enumerator description is assumed. The norm-separation identity also gives an explicit ideal-lattice realization of the exact orthogonal case in the lattice-extension framework of [8, Lemma 4].

The product depends only on the coset $c + qR$, rather than on the chosen representative. Indeed, if $c' = c + q \sum_{j=0}^{\ell-1} r_j b_j$, with $r_j \in R_0$, then $c'_j = c_j + q r_j$ and hence $c'_j + qR_0 = c_j + qR_0$ for every j .

Combining the factorization with the coset decomposition gives the theta series of the entire q -ary ideal.

Corollary 2 (Theta Series of a q -ary Ideal). *Let $I \subseteq R$ be a q -ary ideal, and let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be a norm-separating R_0 -basis of R , with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. For each $c \in I_q$, fix a representative in I and write $c = \sum_{j=0}^{\ell-1} c_j b_j$, $c_j \in R_0$. Then, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$, $\Theta_I^K(\tau) = \sum_{c \in I_q} \prod_{j=0}^{\ell-1} \Theta_{c_j + qR_0}^{K_0}(\lambda_j \tau)$.*

Proof. By Lemma 1, $\Theta_I^K(\tau) = \sum_{c \in I_q} \Theta_{c + qR}^K(\tau)$. Applying Theorem 2 to every coset gives the claimed identity.

At $\tau = i/s^2$, this reduces each within-coset evaluation to lower dimension while leaving the outer sum over I_q explicit. Factors with the same R_0/qR_0 -coset and scaling parameter can be reused; the same separation between within-coset sampling and class selection appears in [4, Theorem 1].

4 Cyclotomic and Non-Cyclotomic Extensions

Section 3 shows that theta-series dimension reduction follows whenever R admits a norm-separating R_0 -basis $\mathbf{b} = (b_0, \dots, b_{\ell-1})$, satisfying

$$\left\| \sigma_K \left(\sum_{j=0}^{\ell-1} a_j b_j \right) \right\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(a_j)\|^2$$

for some $\lambda_0, \dots, \lambda_{\ell-1} \in \mathbb{R}_{>0}$. Under this condition, Theorem 2 gives the coset factorization and, at $\tau = i/s^2$, the corresponding exact within-coset Gaussian decomposition. The trace criterion in Lemma 2 is sufficient but not required.

We first consider cyclotomic extensions. Powerful bases provide structured coordinates for cyclotomic rings and their canonical embeddings in Ring-LWE cryptography [12]. In the context of field switching for homomorphic encryption, Gentry, Halevi, Peikert, and Smart construct relative powerful bases whose

embedding matrices have controlled singular values [10, Sec. 2.1.5 and Lemma 2.6]. Under the condition below, the resulting embedding matrix is scaled unitary. We show that this geometric property yields the relative trace identities required by Lemma 2, with $\lambda_i = \ell = [K : K_0]$ for every i . Consequently, all lower-dimensional Gaussian factors have the same parameter $s/\sqrt{\ell}$.

We also construct two non-cyclotomic families directly from their embeddings. Their scaling factors need not agree, but once the R_0 -module and norm-separation conditions are verified, the corresponding formulas follow from Theorem 2.

4.1 Cyclotomic Extensions

Cyclotomic rings are a standard algebraic setting for Ring-LWE-based cryptography [12]. Let $K = \mathbb{Q}(\zeta_m)$, $K_0 = \mathbb{Q}(\zeta_{m_0})$, $m_0 \mid m$, where $\zeta_{m_0} = \zeta_m^{m/m_0}$. Their rings of integers are $R = \mathbb{Z}[\zeta_m]$ and $R_0 = \mathbb{Z}[\zeta_{m_0}]$. Write $n = \varphi(m)$, $n_0 = \varphi(m_0)$, $\ell = [K : K_0] = n/n_0$.

For a positive integer u , let $\text{rad}(u) := \prod_{p \mid u} p$. The condition $\text{rad}(m)/\text{rad}(m_0) \in \{1, 2\}$ means that no new odd prime divisor appears when passing from m_0 to m . The exponents of primes already dividing m_0 may still increase. Gentry, Halevi, Peikert, and Smart construct an efficiently computable R_0 -basis of R and show that, under this condition, its matrix under the canonical embedding is scaled unitary [10, Sec. 2.1.5 and Lemma 2.6]. This construction is used for field switching in homomorphic encryption. We use the same basis to verify the relative trace condition from Lemma 2. An explicit description of this basis is given in Appendix A.

Proposition 2. *Assume that $\text{rad}(m)/\text{rad}(m_0) \in \{1, 2\}$. Then there exists an efficiently computable R_0 -basis $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ of R such that*

$$\text{Tr}_{K/K_0}(b_i \overline{b_j}) = \begin{cases} \ell, & i = j, \\ 0, & i \neq j. \end{cases}$$

Proof. For an ordered R_0 -basis $\mathbf{b} = (b_0, \dots, b_{\ell-1})$, let $\mathbf{M}_{\mathbf{b}} \in \mathbb{C}^{n \times n}$ be the matrix determined by $\sigma_K \left(\sum_{j=0}^{\ell-1} a_j b_j \right) = \mathbf{M}_{\mathbf{b}} (\sigma_0(a_0), \dots, \sigma_0(a_{\ell-1}))$ for $a_0, \dots, a_{\ell-1} \in K_0$, where the vector on the right is obtained by concatenating the canonical embeddings of the coefficients. Define

$$\widehat{m} := \begin{cases} m/2, & \text{if } m \text{ is even and } m_0 \text{ is odd,} \\ m, & \text{otherwise.} \end{cases}$$

By [10, Lemma 2.6], there exists an efficiently computable R_0 -basis $\mathbf{b}$ for which, under the stated condition, $M_{\mathbf{b}}$ is a unitary matrix scaled by $\sqrt{\widehat{m}/m_0}$. Hence, $\mathbf{M}_{\mathbf{b}}^* \mathbf{M}_{\mathbf{b}} = \widehat{m} \mathbf{I}_n / m_0$. Moreover, under the stated condition, the prime-power factorization of the Euler totient function gives $\widehat{m}/m_0 = \varphi(m)/\varphi(m_0) = \ell$. It follows that, for every $a, b \in K_0$ and $0 \leq i, j < \ell$, $\langle \sigma_K(ab_i), \sigma_K(bb_j) \rangle = \ell \delta_{ij} \langle \sigma_0(a), \sigma_0(b) \rangle$.

Set $t_{ij} := \text{Tr}_{K/K_0}(b_i \bar{b}_j)$. By Proposition 1 and the transitivity of the field trace, the preceding identity is equivalent to $\text{Tr}_{K_0/\mathbb{Q}}(a \bar{b} t_{ij}) = \ell \delta_{ij} \text{Tr}_{K_0/\mathbb{Q}}(a \bar{b})$. Taking $b = 1$ and using the nondegeneracy of the trace pairing on K_0 gives $t_{ij} = \ell \delta_{ij}$, as required.

For a general cyclotomic conductor m , a natural choice is $m_0 = \text{rad}(m)$. Then the assumption of Proposition 2 holds automatically. This choice keeps one copy of each prime divisor in the subfield and places the higher prime powers in the extension K/K_0 . If m is square-free, then $m_0 = m$ and $\ell = 1$, so this choice gives no nontrivial dimension reduction. Other choices of m_0 are possible whenever they satisfy the same radical condition.

Because the relative trace values are all ℓ , Theorem 2 specializes as follows.

Corollary 3 (Theta-Series Decomposition for Cyclotomic Extensions).

Let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be the relative basis from Proposition 2. For $c \in R$, write $c = \sum_{j=0}^{\ell-1} c_j b_j$, $c_j \in R_0$. Then, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$, $\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{\ell-1} \Theta_{c_j+qR_0}^{K_0}(\ell\tau)$.

Moreover, let $I \subseteq R$ be a q -ary ideal. For every $c \in I_q$, fix a representative in I and write $c = \sum_{j=0}^{\ell-1} c_j b_j$. Then

$$\Theta_I^K(\tau) = \sum_{c \in I_q} \prod_{j=0}^{\ell-1} \Theta_{c_j+qR_0}^{K_0}(\ell\tau).$$

Proof. Proposition 2 verifies the hypotheses of Lemma 2 with $\lambda_0 = \dots = \lambda_{\ell-1} = \ell$. The first identity follows from Theorem 2, and the second follows from Corollary 2.

At $\tau = i/s^2$, the factors give ℓ independent within-coset Gaussian distributions, all with parameter $s/\sqrt{\ell}$. The decomposition is exact for every $s > 0$; the next subsection treats families with unequal scaling factors.

4.2 Non-Cyclotomic Extensions

We consider rational radical extensions and maximal real subfields, using explicit relative bases to determine their factors and Gaussian parameters.

Rational Radical Extensions Let K_0 be a number field and let $R_0 \subseteq K_0$ be an order. We adjoin several radicals of rational integers. When K_0 is cyclotomic and all relative degrees are two, this setting includes the cyclo-multiquadratic rings studied for fast polynomial arithmetic and RLWE–PLWE compatibility in homomorphic encryption [3]. The construction below does not restrict the radical degrees r_i to two.

Proposition 3. *Let $t \geq 1$, $r_1, \dots, r_t \geq 2$, and choose $\alpha_1, \dots, \alpha_t$ such that $\alpha_i^{r_i} = d_i$, $d_i \in \mathbb{Z} \setminus \{0\}$. Set $K = K_0(\alpha_1, \dots, \alpha_t)$ and assume that $[K : K_0] = \ell := \prod_{i=1}^t r_i$. Let $R = R_0[\alpha_1, \dots, \alpha_t]$ and define*

$$\mathcal{J} := \{\mathbf{j} = (j_1, \dots, j_t) : 0 \leq j_i < r_i\}, \quad b_{\mathbf{j}} := \prod_{i=1}^t \alpha_i^{j_i}.$$

Then R is an order in K , and $\mathbf{b} = (b_{\mathbf{j}})_{\mathbf{j} \in \mathcal{J}}$ is an R_0 -basis of R . Moreover, for all $a_{\mathbf{j}} \in K_0$,

$$\left\| \sigma_K \left(\sum_{\mathbf{j} \in \mathcal{J}} a_{\mathbf{j}} b_{\mathbf{j}} \right) \right\|^2 = \sum_{\mathbf{j} \in \mathcal{J}} \lambda_{\mathbf{j}} \|\sigma_0(a_{\mathbf{j}})\|^2,$$

where $\lambda_{\mathbf{j}} = \prod_{i=1}^t r_i |d_i|^{2j_i/r_i} = \ell \prod_{i=1}^t |d_i|^{2j_i/r_i}$.

Proof. The elements α_i are algebraic integers. The relations $\alpha_i^{r_i} = d_i$ show that the monomials $b_{\mathbf{j}}$ span R over R_0 , while the degree assumption makes them linearly independent over K_0 . Hence they form an R_0 -basis of R .

Fix an embedding $\sigma : K_0 \hookrightarrow \mathbb{C}$. For each i , choose a root β_i of $X^{r_i} - d_i$, and write $\zeta_{r_i} = e^{2\pi i/r_i}$. By the degree assumption, the ℓ extensions of σ to K are indexed by $\mathbf{u} = (u_1, \dots, u_t) \in \mathcal{J}$, with $\tilde{\sigma}_{\mathbf{u}}(\alpha_i) = \beta_i \zeta_{r_i}^{u_i}$. Therefore,

$$\sum_{\mathbf{u} \in \mathcal{J}} \tilde{\sigma}_{\mathbf{u}}(b_{\mathbf{j}}) \overline{\tilde{\sigma}_{\mathbf{u}}(b_{\mathbf{k}})} = \prod_{i=1}^t \left(\beta_i^{j_i} \overline{\beta_i}^{k_i} \sum_{u_i=0}^{r_i-1} \zeta_{r_i}^{u_i(j_i - k_i)} \right) = \begin{cases} \lambda_{\mathbf{j}}, & \mathbf{j} = \mathbf{k}, \\ 0, & \mathbf{j} \neq \mathbf{k}. \end{cases}$$

Here we used root-of-unity orthogonality and $|\beta_i| = |d_i|^{1/r_i}$. It follows that

$$\sum_{\tilde{\sigma}|\sigma} \left| \tilde{\sigma} \left(\sum_{\mathbf{j} \in \mathcal{J}} a_{\mathbf{j}} b_{\mathbf{j}} \right) \right|^2 = \sum_{\mathbf{j} \in \mathcal{J}} \lambda_{\mathbf{j}} |\sigma(a_{\mathbf{j}})|^2.$$

Summing over all embeddings of K_0 proves the claimed norm identity.

Unlike the cyclotomic construction in Subsection 4.1, the scaling factors in Proposition 3 need not be equal. They may also be irrational, which is why the norm identity is verified directly rather than through the rational trace criterion of Lemma 2.

Corollary 4 (Theta-Series Decomposition for Radical Extensions).

Under the assumptions of Proposition 3, let $c \in R$ and write $c = \sum_{\mathbf{j} \in \mathcal{J}} c_{\mathbf{j}} b_{\mathbf{j}}$, $c_{\mathbf{j}} \in R_0$. Then, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$, $\Theta_{c+qR}^K(\tau) = \prod_{\mathbf{j} \in \mathcal{J}} \Theta_{c_{\mathbf{j}}+qR_0}^{K_0}(\lambda_{\mathbf{j}}\tau)$.

Moreover, let $I \subseteq R$ be a q -ary ideal. For every $c \in I_q$, fix a representative in I , again denoted by c , and write $c = \sum_{\mathbf{j} \in \mathcal{J}} c_{\mathbf{j}} b_{\mathbf{j}}$. Then

$$\Theta_I^K(\tau) = \sum_{c \in I_q} \prod_{\mathbf{j} \in \mathcal{J}} \Theta_{c_{\mathbf{j}}+qR_0}^{K_0}(\lambda_{\mathbf{j}}\tau).$$

Proof. Proposition 3 shows that $\mathbf{b}$ is norm-separating with scaling factors $\lambda_{\mathbf{j}}$. The first identity follows from Theorem 2, and the second follows from Corollary 2.

At $\tau = i/s^2$, the factor indexed by $\mathbf{j}$ has Gaussian parameter $s/\sqrt{\lambda_{\mathbf{j}}}$. Thus, within a fixed coset, the distribution separates into ℓ independent distributions of dimension $[K_0 : \mathbb{Q}]$. When $K_0 = \mathbb{Q}$, all factors are one-dimensional distributions over integer cosets, for which efficient discrete Gaussian samplers are available [1,16].

Maximal Real Subfields A second construction comes from the maximal real subfields of 2-power cyclotomic fields. Although these fields are contained in cyclotomic fields, they are not themselves full cyclotomic fields. Modified Chebyshev bases and DCT-based arithmetic have also been studied for broader families of maximal real cyclotomic subfields in the RLWE-PLWE setting [2]. In the 2-power case, the embedding matrix has an especially simple orthogonality relation.

Let $M = 2^r$, $r \geq 3$, and set $\psi_M := \zeta_M + \zeta_M^{-1}$, $K := \mathbb{Q}(\psi_M)$. Then $N := [K : \mathbb{Q}] = \varphi(M)/2 = 2^{r-2}$. The ring of integers of K is $R = \mathcal{O}_K = \mathbb{Z}[\psi_M]$ [19]. For this construction, we take $K_0 = \mathbb{Q}$, $R_0 = \mathbb{Z}$, $\ell = N$. Define the modified Chebyshev polynomials by $V_0(x) = 1$, $V_j(x) = 2T_j(x/2)$, $j \geq 1$, where T_j is the j -th Chebyshev polynomial of the first kind.

Proposition 4. *For $0 \leq j < N$, set $b_j := V_j(\psi_M)$. Then $\mathbf{b} = (b_0, \dots, b_{N-1})$ is a $\mathbb{Z}$ -basis of R , and*

$$\mathrm{Tr}_{K/\mathbb{Q}}(b_i b_j) = \begin{cases} N, & i = j = 0, \\ 2N, & i = j \geq 1, \\ 0, & i \neq j. \end{cases}$$

Consequently, for all $a_0, \dots, a_{N-1} \in \mathbb{Q}$,

$$\left\| \sigma_K \left(\sum_{j=0}^{N-1} a_j b_j \right) \right\|^2 = N a_0^2 + 2N \sum_{j=1}^{N-1} a_j^2.$$

Proof. For $j \geq 1$, the polynomial $V_j(x)$ is monic with integer coefficients and degree j . Hence the change of basis from $\{1, \psi_M, \dots, \psi_M^{N-1}\}$ to $\{b_0, \dots, b_{N-1}\}$ is unitriangular, so $\mathbf{b}$ is a $\mathbb{Z}$ -basis of R . The real embeddings of K can be indexed by $0 \leq k < N$, with $\sigma_k(\psi_M) = 2 \cos((2k+1)\pi/2N)$. Since $V_j(2 \cos \theta) = 2 \cos(j\theta)$ for $j \geq 1$, we have $\sigma_k(b_0) = 1$, $\sigma_k(b_j) = 2 \cos((2k+1)j\pi/2N)$. The DCT orthogonality relation therefore gives

$$\sum_{k=0}^{N-1} \sigma_k(b_i) \sigma_k(b_j) = \begin{cases} N, & i = j = 0, \\ 2N, & i = j \geq 1, \\ 0, & i \neq j. \end{cases}$$

This is the stated trace identity. Since K is totally real, the same matrix is the Gram matrix of the canonical embedding, and the norm identity follows.

Corollary 5 (Theta-Series Decomposition for Maximal Real Subfields).

Let $c \in R$ and write $c = \sum_{j=0}^{N-1} c_j b_j$, $c_j \in \mathbb{Z}$. Then, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$,

$$\Theta_{c+qR}^K(\tau) = \Theta_{c_0+q\mathbb{Z}}^{\mathbb{Q}}(N\tau) \prod_{j=1}^{N-1} \Theta_{c_j+q\mathbb{Z}}^{\mathbb{Q}}(2N\tau).$$

Moreover, let $I \subseteq R$ be a q -ary ideal. For every $c \in I_q$, fix a representative in I , again denoted by c , and write $c = \sum_{j=0}^{N-1} c_j b_j$. Then

$$\Theta_I^K(\tau) = \sum_{c \in I_q} \Theta_{c_0+q\mathbb{Z}}^{\mathbb{Q}}(N\tau) \prod_{j=1}^{N-1} \Theta_{c_j+q\mathbb{Z}}^{\mathbb{Q}}(2N\tau).$$

Proof. Proposition 4 gives a norm-separating basis with $\lambda_0 = N$, $\lambda_1 = \dots = \lambda_{N-1} = 2N$. The result follows from Theorem 2 and Corollary 2.

At $\tau = i/s^2$, the first factor in Corollary 5 has Gaussian parameter $s/\sqrt{N}$, while the remaining $N-1$ factors have parameter $s/\sqrt{2N}$. Hence the Gaussian distribution within each coset separates into N one-dimensional distributions with only two distinct parameters. For $r = 3$, this construction gives $K = \mathbb{Q}(\sqrt{2})$. As r increases, it gives a family outside the full-cyclotomic setting with unbounded degree.

5 Explicit Cyclotomic Cases

Section 4 identifies cyclotomic and non-cyclotomic extensions for which the theta-series factorization applies. We now return to cyclotomic rings and make the resulting formulas explicit in three representative cases.

For a power-of-two conductor, taking $K_0 = \mathbb{Q}$ reduces each coset theta series to a product of one-dimensional theta series. These factors can also be grouped by Lee weight, giving a symmetrized weight-enumerator formula related to the Construction A description in [4]. For an odd prime conductor, no nontrivial cyclotomic subfield reduction of this form is available, and the remaining theta series is described using the dual root lattice $A_{p-1}^{\vee}$ [8, Section 2.4]. For an odd prime-power conductor p^k , choosing $K_0 = \mathbb{Q}(\zeta_p)$ decomposes the original theta series into p^{k-1} prime-cyclotomic factors, each of dimension $p-1$.

5.1 Power-of-Two Cyclotomic Rings

We first consider the power-of-two cyclotomic rings commonly used in Ring-LWE cryptography [12]. Let $m = 2^r$, $r \geq 2$, and write $n = \varphi(m) = 2^{r-1}$. Set $K = \mathbb{Q}(\zeta_m)$, $R = \mathcal{O}_K = \mathbb{Z}[\zeta_m]$. Since $\Phi_m(X) = X^n + 1$, we have $R \simeq \mathbb{Z}[X]/(X^n + 1)$.

We take $K_0 = \mathbb{Q}$ and $R_0 = \mathbb{Z}$. The standard power basis $\mathbf{b} = (b_0, \dots, b_{n-1}) = (1, \zeta_m, \dots, \zeta_m^{n-1})$ is a $\mathbb{Z}$ -basis of R .

The embeddings of K map ζ_m to ζ_m^{2k+1} , $0 \leq k < n$. Root-of-unity orthogonality gives

$$\sum_{k=0}^{n-1} \zeta_m^{(2k+1)(u-v)} = \begin{cases} n, & u = v, \\ 0, & u \neq v, \end{cases} \quad 0 \leq u, v < n.$$

Consequently, for all $a_0, \dots, a_{n-1} \in \mathbb{Q}$,

$$\left\| \sigma_K \left(\sum_{j=0}^{n-1} a_j b_j \right) \right\|^2 = n \sum_{j=0}^{n-1} a_j^2, \quad \Gamma_{\mathbf{b}} = n \mathbf{I}_n.$$

Thus, the norm-separating basis has n one-dimensional components, all with scaling factor n . Fix $c \in R$ and write $c = \sum_{j=0}^{n-1} c_j b_j$, $c_j \in \mathbb{Z}$. By Theorem 2, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$,

$$\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{n-1} \Theta_{c_j+q\mathbb{Z}}^{\mathbb{Q}}(n\tau).$$

Hence, the theta series of the n -dimensional coset $c + qR$ is a product of n one-dimensional theta series.

Now let $I \subseteq R$ be a q -ary ideal. Reduction modulo q gives $R/qR \simeq \mathbb{Z}_q[X]/(X^n + 1)$, so $I_q = I/qR$ can be viewed as a negacyclic $\mathbb{Z}_q$ -linear code of length n . For every $c \in I_q$, fix a representative $c = \sum_{j=0}^{n-1} c_j b_j$, $c_j \in \{0, \dots, q-1\}$. Combining the fixed-coset formula above with Lemma 1 gives

$$\Theta_I^K(\tau) = \sum_{c \in I_q} \prod_{j=0}^{n-1} \Theta_{c_j+q\mathbb{Z}}^{\mathbb{Q}}(n\tau).$$

Thus, the theta series of the ideal lattice is expressed entirely in terms of one-dimensional theta series. Equal factors may be grouped according to the Lee weights of the coefficients, giving the symmetrized weight-enumerator formulation used in the Construction A setting of [4]. We record this equivalent formulation in Appendix B.

We next consider the prime cyclotomic case, where such a complete product decomposition is no longer available.

5.2 Prime Cyclotomic Rings

Let p be an odd prime, and set $K = \mathbb{Q}(\zeta_p)$, $R = \mathcal{O}_K = \mathbb{Z}[\zeta_p]$. Among the cyclotomic subfields $\mathbb{Q}(\zeta_{m_0})$ with $m_0 \mid p$, the only proper choice is $\mathbb{Q}$. However, $\text{rad}(p)/\text{rad}(1) = p \notin \{1, 2\}$. Therefore, the cyclotomic construction of Subsection 4.1 does not give a nontrivial factorization of the $(p-1)$ -dimensional theta

series. The prime cyclotomic case is instead the basic block at which this subfield reduction stops.

This block has a concrete description in terms of the dual root lattice $A_{p-1}^\vee$ [8, Section 2.4]. Let $\mathbf{1} = (1, \dots, 1)^T \in \mathbb{R}^p$, and let $\pi_{\mathbf{1}^\perp}$ denote the orthogonal projection onto $\mathbf{1}^\perp$. We use the realization $A_{p-1}^\vee = \pi_{\mathbf{1}^\perp}(\mathbb{Z}^p) \subseteq \mathbf{1}^\perp$. For $0 \leq j \leq p-2$, define $v_j = e_{j+1} - (1/p) \cdot \mathbf{1}$. Then $(v_0, \dots, v_{p-2})$ is a $\mathbb{Z}$ -basis of $A_{p-1}^\vee$, and $\langle v_u, v_v \rangle = \delta_{uv} - 1/p$.

The $\mathbb{Z}$ -linear map $\iota_p : R \rightarrow A_{p-1}^\vee$, $\iota_p(\zeta_p^j) = v_j$, $0 \leq j \leq p-2$, is an isomorphism satisfying $\|\sigma_K(x)\|^2 = p\|\iota_p(x)\|^2$ for every $x \in R$ [8, Section 2.4]. Consequently, the Gram matrix of the power basis $\mathcal{B} = (1, \zeta_p, \dots, \zeta_p^{p-2})$ is

$$\mathbf{\Gamma}_B = p\mathbf{I}_{p-1} - \mathbf{J}_{p-1},$$

where $\mathbf{J}_{p-1}$ is the $(p-1) \times (p-1)$ all-ones matrix.

Proposition 5 (Theta Series of Prime Cyclotomic Cosets). *Let $c \in R$ and $q \geq 2$. For every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$,*

$$\Theta_{c+qR}^K(\tau) = \Theta_{\iota_p(c)+qA_{p-1}^\vee}(p\tau).$$

Moreover, let $I \subseteq R$ be a q -ary ideal. For every $c \in I_q$, fix a representative in I , again denoted by c . Then

$$\Theta_I^K(\tau) = \sum_{c \in I_q} \Theta_{\iota_p(c)+qA_{p-1}^\vee}(p\tau).$$

Proof. Since ι_p is a $\mathbb{Z}$ -module isomorphism, $\iota_p(c + qR) = \iota_p(c) + qA_{p-1}^\vee$. The norm identity therefore gives

$$\Theta_{c+qR}^K(\tau) = \sum_{x \in c+qR} \exp(\pi i \tau \|\sigma_K(x)\|^2) = \sum_{\mathbf{v} \in \iota_p(c)+qA_{p-1}^\vee} \exp(\pi i (p\tau) \|\mathbf{v}\|^2),$$

which proves the first identity. Summing this identity over $c \in I_q$ and applying Lemma 1 proves the second.

Although Proposition 5 does not give a product decomposition, the Gram matrix still yields an auxiliary integral representation at purely imaginary arguments. We record this representation in Appendix C. The prime-cyclotomic theta series remains the lower-dimensional building block for the prime-power case in Subsection 5.3.

5.3 Prime-Power Cyclotomic Rings

Let p be an odd prime and $k \geq 2$. Set $K = \mathbb{Q}(\zeta_{p^k})$, $R = \mathcal{O}_K = \mathbb{Z}[\zeta_{p^k}]$. Unlike the prime cyclotomic case, K contains the proper cyclotomic subfield $K_0 = \mathbb{Q}(\zeta_p)$, $R_0 = \mathcal{O}_{K_0} = \mathbb{Z}[\zeta_p]$, where $\zeta_p = \zeta_{p^k}^{p^{k-1}}$. The relative degree is $\ell = [K : K_0] = \varphi(p^k)/\varphi(p) = p^{k-1}$. Moreover, $\text{rad}(p^k)/\text{rad}(p) = 1$, so the cyclotomic construction of Subsection 4.1 applies.

The relative power basis $\mathbf{b} = (b_0, \dots, b_{\ell-1}) = (1, \zeta_{p^k}, \dots, \zeta_{p^k}^{\ell-1})$ is an R_0 -basis of R . It is the prime-power instance of the relative powerful bases used for cyclotomic field switching [10, Section 2.1.5 and Lemma 2.6]. Root-of-unity orthogonality gives $\text{Tr}_{K/K_0}(b_u \overline{b_v}) = \ell \delta_{uv}$, $0 \leq u, v < \ell$. Thus, all relative components have the same scaling factor $\ell = p^{k-1}$.

Let $\mathcal{A} = (1, \zeta_p, \dots, \zeta_p^{p-2})$ be the power basis of R_0 . By Subsection 5.2, its Gram matrix is $\Gamma_p = p\mathbf{I}_{p-1} - \mathbf{J}_{p-1}$. Combining $\mathcal{A}$ with $\mathbf{b}$ gives a $\mathbb{Z}$ -basis $\mathcal{B}$ of R with Gram matrix

$$\Gamma_{\mathcal{B}} = \begin{pmatrix} p^{k-1}\Gamma_p & & & \mathbf{0} \\ & p^{k-1}\Gamma_p & & \\ & & \ddots & \\ \mathbf{0} & & & p^{k-1}\Gamma_p \end{pmatrix},$$

Hence, the $\varphi(p^k) = p^{k-1}(p-1)$ coordinates separate into p^{k-1} mutually orthogonal blocks, each of dimension $p-1$.

Proposition 6 (Theta-Series Decomposition for Prime-Power Cyclotomic Rings). *Let $c \in R$ and write $c = \sum_{j=0}^{p^{k-1}-1} c_j b_j$, $c_j \in R_0$. Then, for every $\tau \in \mathbb{C}$ with $\text{Im}(\tau) > 0$,*

$$\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{p^{k-1}-1} \Theta_{c_j+qR_0}^{K_0}(p^{k-1}\tau).$$

Moreover, if $I \subseteq R$ is a q -ary ideal, then

$$\Theta_I^K(\tau) = \sum_{c \in I_q} \prod_{j=0}^{p^{k-1}-1} \Theta_{c_j+qR_0}^{K_0}(p^{k-1}\tau),$$

where, for each $c \in I_q$, we fix a representative in I and its relative coordinates.

Proof. The relative basis $\mathbf{b}$ is norm-separating with $\lambda_0 = \dots = \lambda_{\ell-1} = \ell = p^{k-1}$. The first identity follows from Theorem 2. Summing over $c \in I_q$ and applying Lemma 1 gives the second.

Each factor in Proposition 6 is a prime cyclotomic coset theta series. Applying Proposition 5 to each factor gives the equivalent expression

$$\Theta_{c+qR}^K(\tau) = \prod_{j=0}^{p^{k-1}-1} \Theta_{\iota_p(c_j)+qA_{p-1}^\vee}^{K_0}(p^k\tau).$$

Thus, the original $\varphi(p^k)$ -dimensional theta series is reduced to p^{k-1} factors of dimension $p-1$. Decompositions into prime-cyclotomic blocks also appear in the smooth-conductor setting of [8, Section 7.2]. The result above applies to every odd prime power p^k .

Together, the three cases yield one-dimensional, single-block, or multiple-block reductions for power-of-two, prime, and prime-power conductors, respectively.

6 Applications

The preceding sections show how the theta series of an ideal lattice can be expressed through lower-dimensional factors. We now apply this decomposition to Gaussian masses, discrete Gaussian sampling, and smoothing parameters, which play central roles in lattice-based cryptography [15,11,12].

Evaluating a theta series at $\tau = i/s^2$ gives the corresponding Gaussian mass and hence the normalizing constant of the discrete Gaussian distribution. For a q -ary ideal, the coset masses also determine the probabilities used to select a class in I_q , as in code-based Gaussian sampling [4]. The same norm-separating structure then gives an exact decomposition of the Gaussian distribution within the selected coset. Finally, Poisson summation relates the theta series of an ideal lattice to the smoothing parameter of its dual. In all three applications, the factorization reduces within-coset evaluations while the outer sum over I_q remains explicit.

Subsection 6.1 considers Gaussian masses and coset weights. Subsection 6.2 derives the corresponding sampling reduction, and Subsection 6.3 applies the same theta-series formulas to smoothing parameters. Implementation details and experimental results are deferred to Section 7.

6.1 Gaussian Masses

Let $I \subseteq R$ be a q -ary ideal, and let $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ be a norm-separating R_0 -basis of R with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. For every $c \in I_q$, fix a representative in I and write $c = \sum_{j=0}^{\ell-1} c_j b_j$, $c_j \in R_0$. Since Gaussian mass is obtained by evaluating the theta series at $\tau = i/s^2$, Theorem 2 gives, for every $s > 0$ and $c \in I_q$,

$$\rho_s(\sigma_K(c + qR)) = \prod_{j=0}^{\ell-1} \rho_{s/\sqrt{\lambda_j}}(\sigma_0(c_j + qR_0)). \quad (1)$$

Summing over $c \in I_q$ gives $\rho_s(\sigma_K(I)) = \sum_{c \in I_q} \prod_{j=0}^{\ell-1} \rho_{s/\sqrt{\lambda_j}}(\sigma_0(c_j + qR_0))$.

The mass of each coset determines its probability under the discrete Gaussian distribution over $\sigma_K(I)$. For $s > 0$, define

$$\mathcal{P}_{I_q,s}(c) := \frac{\rho_s(\sigma_K(c + qR))}{\rho_s(\sigma_K(I))}, \quad c \in I_q.$$

If $X \leftarrow D_{\sigma_K(I),s}$, then $\mathcal{P}_{I_q,s}(c) = \Pr[X \in \sigma_K(c + qR)]$. Coset weights of this form also underlie Gaussian sampling for lattice codes and q -ary Construction A lattices [5, Propositions 1–2] [4, Theorem 1].

The factorization in (1) replaces one n -dimensional coset-mass evaluation by ℓ evaluations of dimension n_0 . These evaluations can be performed independently, and repeated factors can be reused. The number of cosets, however, remains $|I_q| = [I : qR] = q^n/[R : I]$, which may be exponential in n . Thus, the factorization reduces the dimension of each coset-mass evaluation but does

not by itself give an efficient method for evaluating or sampling from the full distribution $\mathcal{P}_{I_q, s}$.

These coset probabilities provide the first stage of the Gaussian sampling procedure considered in the next subsection.

6.2 Gaussian Sampling

After a class $c \in I_q$ has been selected according to $\mathcal{P}_{I_q, s}$, it remains to sample from $D_{\sigma_K(c+qR), s}$. Combining the norm identity of Definition 13 with (1) gives the following exact factorization.

Corollary 6 (Gaussian Decomposition within a Coset). *Under the notation and assumptions of Subsection 6.1, fix $c \in I_q$ and write $c = \sum_{j=0}^{\ell-1} c_j b_j$. For $y_j \in c_j + qR_0$, set $y = \sum_{j=0}^{\ell-1} y_j b_j \in c + qR$. Then, for every $s > 0$,*

$$D_{\sigma_K(c+qR), s}(\sigma_K(y)) = \prod_{j=0}^{\ell-1} D_{\sigma_0(c_j+qR_0), s/\sqrt{\lambda_j}}(\sigma_0(y_j)).$$

Proof. By the norm-separation identity, the Gaussian weight of y is the product of the Gaussian weights of the y_j with parameters $s/\sqrt{\lambda_j}$. Equation (1) gives the same factorization for the normalizing constant. Dividing the two identities proves the claim.

Theorem 3 (Gaussian Sampling Reduction). *Under the notation of Subsection 6.1, an exact sample from $D_{\sigma_K(I), s}$ can be generated as follows:*

1. Sample $c \in I_q$ according to $\mathcal{P}_{I_q, s}$.
2. Write the fixed representative of c as $c = \sum_{j=0}^{\ell-1} c_j b_j$. For each $0 \leq j < \ell$, independently sample

$$\sigma_0(Y_j) \leftarrow D_{\sigma_0(c_j+qR_0), s/\sqrt{\lambda_j}}, \quad Y_j \in c_j + qR_0.$$

3. Output $\sigma_K\left(\sum_{j=0}^{\ell-1} Y_j b_j\right)$.

Proof. By Corollary 6, conditioned on the choice of c , Steps 2 and 3 produce an exact sample from $D_{\sigma_K(c+qR), s}$. Step 1 selects the coset with probability $\mathcal{P}_{I_q, s}(c)$. Therefore, for every $y \in I$, if c is the residue class of y modulo qR , the probability that the algorithm outputs $\sigma_K(y)$ is

$$\mathcal{P}_{I_q, s}(c) D_{\sigma_K(c+qR), s}(\sigma_K(y)) = \frac{\rho_s(\sigma_K(y))}{\rho_s(\sigma_K(I))}.$$

Hence the output distribution is exactly $D_{\sigma_K(I), s}$.

General lattice Gaussian samplers handle arbitrary lattices under conditions determined by the Gaussian parameter and the available basis [11,17]. Here, Corollary 6 uses the additional ideal-lattice structure to replace sampling within a fixed n -dimensional coset by ℓ independent samplings of dimension n_0 .

The decomposition is exact for every $s > 0$ because the relative components are orthogonal. The same orthogonality mechanism gives the exact projected distribution in [8, Lemma 4], whereas [8, Lemma 3] obtains an approximate projected distribution for arbitrary nested lattices under a smoothing condition.

As noted above, the reduction does not by itself make Step 1 efficient: a direct method may still evaluate all $|I_q|$ coset weights. The implementation of fixed-coset computations is considered in Section 7; class selection remains outside the scope of the experiments.

We next turn to smoothing parameters, which are defined through the Gaussian mass of the dual lattice.

6.3 Smoothing Parameters

The smoothing parameter is an important scale in lattice-based cryptography. It appears in worst-case-to-average-case reductions and in the analysis of Gaussian samplers [15,11,12]. For a full-rank lattice $\Lambda \subseteq H_K$ and $0 < \varepsilon < 1$,

$$\rho_{1/s}(\Lambda^* \setminus \{\mathbf{0}\}) = \Theta_{\Lambda^*}(is^2) - 1.$$

Since the function on the right is continuous and strictly decreasing from infinity to zero, $\eta_\varepsilon(\Lambda)$ is the unique positive solution of

$$\Theta_{\Lambda^*}(is^2) - 1 = \varepsilon. \quad (2)$$

We apply this characterization to an ideal lattice. For the determinant formula below, we specialize to the maximal order $R = O_K$. Let $I \subseteq O_K$ be a nonzero ideal and write $\Lambda_I := \sigma_K(I)$. Under the full canonical embedding, $\det(\Lambda_I) = N(I)\sqrt{|\Delta_K|}$. Poisson summation gives $\Theta_I(i/s^2) = \frac{s^n}{\det(\Lambda_I)}\Theta_{\Lambda_I^*}(is^2)$. Consequently, $s = \eta_\varepsilon(\Lambda_I)$ is the unique positive solution of

$$\Theta_I(i/s^2) = \frac{s^n}{N(I)\sqrt{|\Delta_K|}}(1 + \varepsilon). \quad (3)$$

Thus, the smoothing parameter can be characterized using the theta series of the original ideal lattice, without explicitly evaluating the theta series of its dual.

Fix $q \geq 2$ such that $qR \subseteq I$, and let $K_0 \subseteq K$, $R_0 \subseteq R$, and $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ satisfy the hypotheses of Theorem 2, with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. For each $c \in I_q$, use the fixed representative from Subsection 6.1 and write $c = \sum_{j=0}^{\ell-1} c_j b_j$, $c_j \in R_0$. The coset decomposition and Theorem 2 give, for $\text{Im}(\tau) > 0$,

$$\Theta_I(\tau) = \sum_{c \in I_q} \prod_{j=0}^{\ell-1} \Theta_{c_j + qR_0}^{K_0}(\lambda_j \tau). \quad (4)$$

Corollary 7 (Smoothing Parameter from Coset Theta Series). *Under the assumptions above, $\eta_\varepsilon(\Lambda_I)$ is the unique positive solution s of*

$$\sum_{c \in I_q} \prod_{j=0}^{\ell-1} \Theta_{c_j + qR_0}^{K_0} \left(\frac{i\lambda_j}{s^2} \right) = \frac{s^n}{N(I)\sqrt{|\Delta_K|}} (1 + \varepsilon).$$

Proof. Evaluating (4) at $\tau = i/s^2$ shows that the left-hand side is $\Theta_I(i/s^2)$. Equation (3) then shows that the displayed equation is equivalent to $\Theta_{\Lambda_I^*}(is^2) - 1 = \varepsilon$. By (2), its unique positive solution is $s = \eta_\varepsilon(\Lambda_I)$.

The corollary replaces each n -dimensional coset theta series by ℓ factors of dimension $n_0 = n/\ell$, while the outer sum and its possible dependence on $|I_q|$ remain as described above. The complexity of approximating smoothing parameters for general lattices was studied in [6].

7 Implementation and Experiments

The preceding sections establish exact factorizations for the theta series and Gaussian distributions of ideal-lattice cosets. This section evaluates their practical effect on theta-series evaluation, numerical smoothing-parameter computation, and Gaussian sampling within a fixed coset.

The experiments use the cyclotomic instances described in Section 4. The theta-series and sampling experiments measure computations after a coset has been fixed; selecting a class from I_q remains outside their scope. We also measure the numerical solution of the smoothing-parameter equation below.

7.1 Implementation and Methodology

For each fixed $c \in I_q$, the implementation constructs the relative basis, writes $c = \sum_{j=0}^{\ell-1} c_j b_j$, evaluates the required lower-dimensional factors, and recombines the outputs in the basis $(b_0, \dots, b_{\ell-1})$.

The tested conductors have the form $m = 2^a p^k$, $p \in \{3, 5, 7\}$, $k \in \{1, 2\}$. Unless stated otherwise, we use $K_0 = \mathbb{Q}(\zeta_p)$, $n_0 = p - 1$, $\ell = n/(p - 1)$. The main sampling experiments use $q = 2$. Additional tests use $q \in \{2, 4, 8\}$, zero and nonzero cosets, and Gaussian parameters in $\{1.5, 2, 2.5, 3, 4\}$.

For Gaussian sampling, both routes use the GPV sampler provided by SageMath [11, 18]. Matching the convention $\exp(-\pi\|\mathbf{x} - \mathbf{c}\|^2/s^2)$ used in this paper with SageMath's convention $\exp(-\|\mathbf{x} - \mathbf{c}\|^2/(2\sigma^2))$ gives $\sigma = s/\sqrt{2\pi}$. For a reduced block with scaling factor λ_j , the corresponding parameter is $\sigma_j = s/\sqrt{2\pi\lambda_j}$.

For theta-series evaluation, a Fincke–Pohst-style enumeration [9] is used with the same numerical stopping target 0.002 for all routes. This target is a numerical stopping rule, not a certified bound on the remaining tail. Theta-series times are averaged over three runs, sampling times are reported per sample, and smoothing-parameter times use three repetitions and report medians. The speedups in the tables are computed relative to the corresponding baseline implementation.

7.2 Theta Series of a Fixed Coset

Theorem 2 evaluates the theta series of a fixed coset as a product of lower-dimensional factors. We compare this factored route with two full-dimensional routes. Full-PB uses the original power basis, while Full-RB uses the relative basis but still performs one full-dimensional enumeration. Factored evaluates the lower-dimensional factors independently and then multiplies them.

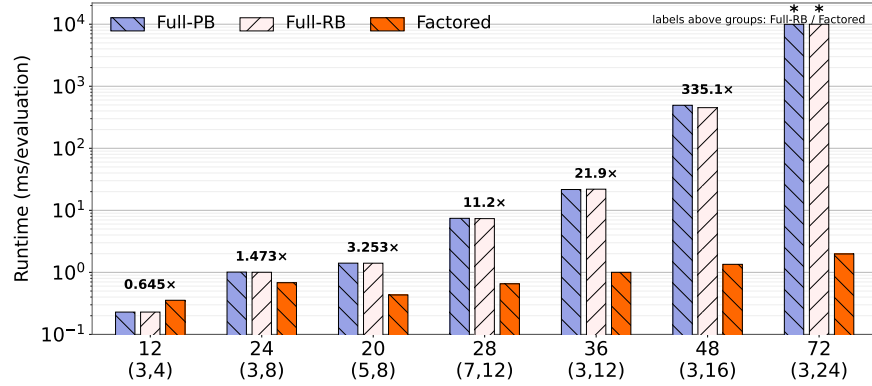


Fig. 1: Theta-series evaluation times. The horizontal axis shows m and (m_0, n) , where m_0 defines the subfield and $n = \varphi(m)$. Numbers above the bars show Full-RB/Factored. An asterisk means the enumeration reached its budget of 10^6 solutions before meeting the stopping target.

As shown in Figure 1, Full-PB and Full-RB have nearly identical running times on the smaller instances, indicating that changing the basis alone provides little benefit. For $m = 12$, both take 0.229 ms, while Factored takes 0.355 ms. The advantage of the product formula becomes clearer on larger instances: for $m = 48$ ($n = 16$), Factored reduces the runtime from 451.786 ms for Full-RB to 1.348 ms, a speedup of approximately 335 $\times$. For the largest instance, $m = 72$ ($n = 24$), both full-dimensional methods exhaust the 10^6 -solution budget without meeting the stopping target, whereas Factored completes the evaluation in 1.979 ms.

7.3 Smoothing-Parameter Evaluation

We also measure the numerical solution of the smoothing-parameter equation using the dual theta-series evaluators. The full power-basis and full relative-basis routes enumerate the complete dual lattice, whereas the factored route evaluates one low-dimensional dual block and applies the exact identical-block product. Each route was run three times for the six cyclotomic instances below and $\varepsilon \in \{2^{-20}, 2^{-40}, 2^{-64}\}$. Table 1 reports the median total-time speedup of the full relative route over the factored route, including evaluator setup. The factored route is faster at all 18 parameter points, with speedups from 1.09 $\times$ to 215.45 $\times$.

Table 1: End-to-end smoothing-parameter times (ms) and speedups. Each speedup is the full relative-basis time divided by the factored time, including evaluator setup.

Instance		$\varepsilon = 2^{-20}$			$\varepsilon = 2^{-40}$			$\varepsilon = 2^{-64}$		
m	n	Full	Fact.	Spd.	Full	Fact.	Spd.	Full	Fact.	Spd.
12	4	2.594	1.889	1.373×	2.523	1.762	1.431×	1.791	1.643	1.090×
20	8	10.904	2.111	5.165×	11.033	2.255	4.893×	10.279	2.710	3.794×
24	8	11.790	1.952	6.039×	7.441	1.828	4.071×	7.134	2.149	3.320×
28	12	4.643	2.672	1.738×	226.237	2.494	90.715×	224.686	4.793	46.881×
36	12	197.279	1.683	117.227×	3.553	1.661	2.139×	3.357	1.551	2.165×
48	16	323.487	1.666	194.206×	358.376	1.663	215.451×	4.651	1.694	2.745×

Speedups vary nonmonotonically because the Fincke–Pohst workload changes discretely with the cutoff. For $(m, n) = (36, 12)$ and $\varepsilon = 2^{-40}$, the full-relative route enumerates only 1,254 nodes, so setup and fixed overheads dominate and the speedup is 2.139×. At $\varepsilon = 2^{-20}$, the same route enumerates 48,966 nodes and yields 117.227× speedup. This variation reflects the enumeration workload; the factorization remains exact.

7.4 Gaussian Sampling from a Fixed Coset

Corollary 6 replaces one n -dimensional sample from $c + qR$ by ℓ independent samples from the lower-dimensional cosets $c_j + qR_0$, followed by a deterministic recombination. The Direct route invokes the same SageMath sampler once on the complete coset, while the Reduced route uses the factorization.

Table 2: Sequential Gaussian sampling from a fixed coset using the same SageMath backend. Here $n = \varphi(m)$, and times are in milliseconds per sample.

Instance			Runtime (ms/sample)		Speedup
m	m_0	n	Direct	Reduced	
36	3	12	0.728	1.369	0.532×
576	3	192	25.763	19.503	1.321×
1152	3	384	117.737	38.825	3.033×
2304	3	768	476.689	78.182	6.097×
200	5	80	6.596	6.155	1.072×
800	5	320	72.296	24.366	2.967×
1600	5	640	325.539	49.086	6.632×
196	7	84	7.155	5.766	1.241×
784	7	336	77.483	23.051	3.361×
1568	7	672	366.130	46.206	7.924×

As shown in Table 2, the Reduced route can be slower for the smallest instance because the additional sampler calls and recombination dominate. For larger dimensions, the lower-dimensional computations become advantageous and the reduced route consistently improves on the direct computation.

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Supplementary Material

A Explicit Cyclotomic Relative Basis

We use the notation of Subsection 4.1 and make explicit the R_0 -basis used in Proposition 2. Write $m = \prod_{r=1}^t p_r^{e_r}$, $m_0 = \prod_{r=1}^t p_r^{e_{0,r}}$, where $p_1, \dots, p_t$ are distinct primes and $0 \leq e_{0,r} \leq e_r$. Set $m_r := p_r^{e_r}$, $m_{0,r} := p_r^{e_{0,r}}$, with $p_r^0 = 1$, and regard $\zeta_{m_r} := \zeta_m^{m/m_r}$ as an element of $R = \mathbb{Z}[\zeta_m]$. For each $1 \leq r \leq t$, define

$$\mathcal{B}_r := \begin{cases} \left\{ 1, \zeta_{m_r}, \dots, \zeta_{m_r}^{m_r/m_{0,r}-1} \right\}, & e_{0,r} > 0, \\ \left\{ 1, \zeta_{m_r}, \dots, \zeta_{m_r}^{\varphi(m_r)-1} \right\}, & e_{0,r} = 0. \end{cases}$$

The relative-basis construction of [10, Section 2.1.5] shows that

$$\mathcal{B} := \left\{ \prod_{r=1}^t \beta_r : \beta_r \in \mathcal{B}_r \text{ for every } r \right\}$$

is an R_0 -basis of R . Its cardinality is

$$|\mathcal{B}_r| = \begin{cases} \frac{m_r}{m_{0,r}} = \frac{\varphi(m_r)}{\varphi(m_{0,r})}, & e_{0,r} > 0, \\ \varphi(m_r), & e_{0,r} = 0, \end{cases}$$

and hence

$$|\mathcal{B}| = \prod_{r=1}^t |\mathcal{B}_r| = \frac{\varphi(m)}{\varphi(m_0)} = [K : K_0] = \ell.$$

Choosing any fixed ordering of $\mathcal{B}$ gives the ordered basis $\mathbf{b} = (b_0, \dots, b_{\ell-1})$. Under the radical condition of Proposition 2, its embedding matrix has the scaled-unitary property proved in [10, Lemma 2.6]. Thus, this is precisely the basis used to obtain the relative trace identity in that proposition.

B Lee-Weight Form of the Power-of-Two Case

We use the notation of Subsection 5.1. For $a \in \mathbb{Z}_q$, represented by an integer in $\{0, \dots, q-1\}$, define its Lee weight by $w_{\text{Lee}}(a) := \min\{a, q-a\}$. Since $\Theta_{a+q\mathbb{Z}}^{\mathbb{Q}}(\tau) = \Theta_{-a+q\mathbb{Z}}^{\mathbb{Q}}(\tau)$, the one-dimensional theta factor associated with a depends only on $w_{\text{Lee}}(a)$. For $c = (c_0, \dots, c_{n-1}) \in I_q$ and $0 \leq i \leq \lfloor q/2 \rfloor$, let $n_i(c) := \#\{j : w_{\text{Lee}}(c_j) = i\}$. Then

$$\Theta_{c+qR}^K(\tau) = \prod_{i=0}^{\lfloor q/2 \rfloor} \left(\Theta_{i+q\mathbb{Z}}^{\mathbb{Q}}(n\tau) \right)^{n_i(c)}.$$

Definition 14 (Symmetrized Weight Enumerator). *The symmetrized weight enumerator of I_q is*

$$\text{swe}_{I_q}(X_0, \dots, X_{\lfloor q/2 \rfloor}) := \sum_{c \in I_q} \prod_{i=0}^{\lfloor q/2 \rfloor} X_i^{n_i(c)}.$$

For each $c \in I_q$, the corresponding monomial records the multiplicity of every one-dimensional theta factor in $\Theta_{c+qR}^K(\tau)$. Therefore, the substitution $X_i = \Theta_{i+q\mathbb{Z}}^{\mathbb{Q}}(n\tau)$ turns this monomial into the theta series of the corresponding coset. Summing over all $c \in I_q$ gives the following Construction A expression [4]:

$$\Theta_I^K(\tau) = \text{swe}_{I_q} \left(\Theta_{q\mathbb{Z}}^{\mathbb{Q}}(n\tau), \Theta_{1+q\mathbb{Z}}^{\mathbb{Q}}(n\tau), \dots, \Theta_{\lfloor q/2 \rfloor + q\mathbb{Z}}^{\mathbb{Q}}(n\tau) \right).$$

C An Integral Representation for Prime Cyclotomic Cosets

We use the notation of Subsection 5.2. Let p be an odd prime, let $q \geq 2$, and write $c = \sum_{j=0}^{p-2} c_j \zeta_p^j$, $c_j \in \mathbb{Z}$. For $z = (z_0, \dots, z_{p-2}) \in \mathbb{Z}^{p-1}$, set $y_j := c_j + qz_j$.

Proposition 7 (Integral Representation of a Prime-Cyclotomic Coset).
For every $s > 0$,

$$\Theta_{c+qR}^K(i/s^2) = \int_{\mathbb{R}} e^{-\pi u^2} \prod_{j=0}^{p-2} F_{c_j}(u; s) \, du,$$

where

$$F_a(u; s) := \sum_{z \in \mathbb{Z}} \exp \left(-\frac{\pi p}{s^2} (a + qz)^2 + \frac{2\pi u}{s} (a + qz) \right).$$

Proof. Let $\mathbf{y} = (y_0, \dots, y_{p-2})^T$. By the Gram-matrix identity $\mathbf{\Gamma}_{\mathcal{B}} = p\mathbf{I}_{p-1} - \mathbf{J}_{p-1}$, the squared canonical norm of the corresponding element of $c + qR$ is

$$\|\sigma_K(c + qz)\|^2 = \mathbf{y}^T (p\mathbf{I}_{p-1} - \mathbf{J}_{p-1}) \mathbf{y} = p \sum_{j=0}^{p-2} y_j^2 - \left(\sum_{j=0}^{p-2} y_j \right)^2.$$

Substituting this identity into the definition of the theta series at $\tau = i/s^2$ gives the first formula.

Next, use the Gaussian integral identity

$$e^{\pi t^2} = \int_{\mathbb{R}} e^{-\pi u^2 + 2\pi ut} \, du$$

with $t = (\sum_{j=0}^{p-2} y_j)/s$. The resulting summands are nonnegative, so Tonelli's theorem allows us to interchange the sum and the integral. The sum then separates

over the coordinates $y_j = c_j + qz_j$, which gives

$$\Theta_{c+qR}^K(i/s^2) = \int_{\mathbb{R}} e^{-\pi u^2} \prod_{j=0}^{p-2} \left(\sum_{z_j \in \mathbb{Z}} \exp \left(-\frac{\pi p}{s^2} (c_j + qz_j)^2 + \frac{2\pi u}{s} (c_j + qz_j) \right) \right) du.$$

This is the stated integral representation.

The integrand is a product of one-dimensional sums, but the integration variable u is shared by all factors. Thus, this representation does not give a product decomposition of the prime-cyclotomic coset theta series. It is included as an auxiliary formula. The actual product decomposition used later comes from the relative-field construction in Subsection 5.3.

D A Finite Test for Norm-Separating Relative Bases

We give an exact finite test for the existence of a norm-separating relative basis. Let $\mathcal{A} = (a_1, \dots, a_{n_0})$ be a $\mathbb{Z}$ -basis of R_0 , and let $\mathcal{E} = (e_1, \dots, e_n)$ be a $\mathbb{Z}$ -basis of R . Denote their canonical Gram matrices by

$$\Gamma_{\mathcal{A}} = (\langle \sigma_0(a_u), \sigma_0(a_v) \rangle)_{1 \leq u, v \leq n_0}, \quad \Gamma_{\mathcal{E}} = (\langle \sigma_K(e_u), \sigma_K(e_v) \rangle)_{1 \leq u, v \leq n}.$$

Define

$$P := \left(\frac{\det(\Gamma_{\mathcal{E}})}{(\det(\Gamma_{\mathcal{A}}))^{\ell}} \right)^{1/n_0}, \quad T := \frac{n_0 P}{\ell^{\ell-1}}.$$

Proposition 8 (Finite Existence Test). *There exists a norm-separating R_0 -basis of R if and only if R is free of rank ℓ as an R_0 -module and there is an ordered tuple $\mathbf{b} = (b_0, \dots, b_{\ell-1})$ with $b_j \in R \setminus \{0\}$, $\|\sigma_K(b_j)\|^2 \leq T$, such that the following two conditions hold.*

First, the elements $a_1 b_0, \dots, a_{n_0} b_0, a_1 b_1, \dots, a_{n_0} b_1, \dots, a_1 b_{\ell-1}, \dots, a_{n_0} b_{\ell-1}$ form a $\mathbb{Z}$ -basis of R . Second, if $\Gamma_{\mathbf{b}}$ denotes the canonical Gram matrix of this ordered basis, then

$$\Gamma_{\mathbf{b}} = \begin{pmatrix} \lambda_0 \Gamma_{\mathcal{A}} & & & \mathbf{0} \\ & \lambda_1 \Gamma_{\mathcal{A}} & & \\ & & \ddots & \\ \mathbf{0} & & & \lambda_{\ell-1} \Gamma_{\mathcal{A}} \end{pmatrix},$$

where $\lambda_j = \|\sigma_K(b_j)\|^2 / n_0$.

Proof. Assume first that $\mathbf{b}$ is a norm-separating R_0 -basis with scaling factors $\lambda_0, \dots, \lambda_{\ell-1}$. Since every nonzero element of R is an algebraic integer, the arithmetic-geometric mean inequality gives

$$\|\sigma_K(b_j)\|^2 = \sum_{\sigma: K \hookrightarrow \mathbb{C}} |\sigma(b_j)|^2 \geq n |N_{K/\mathbb{Q}}(b_j)|^{2/n} \geq n.$$

Hence $\lambda_j = \|\sigma_K(b_j)\|^2/n_0 \geq \ell$.

The ordered $\mathbb{Z}$ -basis obtained by combining $\mathcal{A}$ with $\mathbf{b}$ has Gram matrix $\text{diag}(\lambda_0 \mathbf{\Gamma}_{\mathcal{A}}, \dots, \lambda_{\ell-1} \mathbf{\Gamma}_{\mathcal{A}})$. Its determinant is also $\det(\mathbf{\Gamma}_{\mathcal{E}})$, since both ordered sets are $\mathbb{Z}$ -bases of R . Therefore, $\det(\mathbf{\Gamma}_{\mathcal{E}}) = (\det(\mathbf{\Gamma}_{\mathcal{A}}))^\ell \prod_{j=0}^{\ell-1} \lambda_j^{n_0}$, and consequently $P = \prod_{j=0}^{\ell-1} \lambda_j$. Since every $\lambda_j \geq \ell$, $\lambda_j \leq P/\ell^{\ell-1}$. It follows that

$$\|\sigma_K(b_j)\|^2 = n_0 \lambda_j \leq \frac{n_0 P}{\ell^{\ell-1}} = T.$$

Thus every norm-separating basis occurs among the tuples considered in the proposition.

Conversely, suppose that a tuple $\mathbf{b}$ satisfies the two stated conditions. The first condition shows that $\mathbf{b}$ is an R_0 -basis of R . The block-diagonal Gram identity gives, for all $a, b \in R_0$,

$$\langle \sigma_K(ab_i), \sigma_K(bb_j) \rangle = \begin{cases} \lambda_i \langle \sigma_0(a), \sigma_0(b) \rangle, & i = j, \\ 0, & i \neq j. \end{cases}$$

Since R_0 spans K_0 over $\mathbb{Q}$, this identity extends by $\mathbb{Q}$ -bilinearity to all $a, b \in K_0$. Therefore, for all $a_0, \dots, a_{\ell-1} \in K_0$,

$$\left\| \sigma_K \left(\sum_{j=0}^{\ell-1} a_j b_j \right) \right\|^2 = \sum_{j=0}^{\ell-1} \lambda_j \|\sigma_0(a_j)\|^2.$$

Hence $\mathbf{b}$ is norm-separating.

It is worth noting that the test is finite because R is a lattice in H_K , so it contains only finitely many elements in the ball $\{x \in R : \|\sigma_K(x)\|^2 \leq T\}$. Concretely, if $\mathbf{z} \in \mathbb{Z}^n$ is the coordinate vector of x with respect to $\mathcal{E}$, then $\mathbf{z}^T \mathbf{\Gamma}_{\mathcal{E}} \mathbf{z} \leq T$. Thus the candidate elements can be obtained by enumerating the integer points in this ellipsoid. For every ordered ℓ -tuple of candidates, the basis condition is checked by the determinant of its coordinate matrix, and the norm-separation condition is checked by the displayed block-diagonal Gram identity.

This gives an exact existence decision procedure. The enumeration is not intended as an efficient method for cryptographic parameter sizes. Its purpose is to distinguish the existence question from the verification of a basis that has already been constructed.

Remark 1. Under the complex-conjugation compatibility assumption of Lemma 2, the enumeration can be preceded by a simple obstruction test. If a norm-separating basis exists, then it can be normalized so that $b_0 = 1$, $\lambda_0 = \ell$, and $\text{Tr}_{K/K_0}(R) = \ell R_0$. Therefore, if this equality fails, no norm-separating relative basis exists. For example, for $K = \mathbb{Q}(\zeta_p)$, $K_0 = \mathbb{Q}$, and $R = \mathbb{Z}[\zeta_p]$, one has $\text{Tr}_{K/\mathbb{Q}}(\zeta_p) = -1$, so $\text{Tr}_{K/\mathbb{Q}}(R) = \mathbb{Z} \neq (p-1)\mathbb{Z}$. The prime cyclotomic case is therefore rejected before the finite enumeration.

E Additional Block-Dimension Results

Choice of the subfield. The relative factorization allows a tradeoff between fewer larger blocks and more smaller blocks. Table 3 reports this tradeoff for $m = 800$, $n = 320$, and $q = 2$.

Table 3: Effect of the block dimension for $m = 800$, $n = 320$, and $q = 2$. Times are in milliseconds per sample, and speedup is relative to $n_0 = 320$.

Block dimension		Performance	
n_0	$\ell = n/n_0$	Time	Speedup
4	80	26.353	$2.61\times$
8	40	21.792	$3.16\times$
16	20	19.865	$3.46\times$
20	16	19.756	$3.48\times$
40	8	20.818	$3.31\times$
80	4	26.208	$2.63\times$
160	2	37.814	$1.82\times$
320	1	68.829	$1.00\times$

As shown in Table 3, intermediate block dimensions give the best observed performance. The smallest available block is not necessarily optimal, since smaller blocks require more sampler calls and more recombination work.

F Additional Parallel Sampling Results

The independent lower-dimensional samples can be evaluated concurrently. The following measurement uses $n = 320$, $n_0 = 4$, and $\ell = 80$.

Table 4: Parallel Gaussian sampling for $n = 320$, $n_0 = 4$, and $\ell = 80$. Times are in milliseconds per sample. Speedup is relative to one process.

Processes	Time	Speedup	Parallel efficiency
1	1.372	$1.000\times$	100.0%
2	1.134	$1.210\times$	60.5%
4	1.087	$1.262\times$	31.6%
8	1.048	$1.309\times$	16.4%
16	0.990	$1.386\times$	8.7%

Increasing the process count reduces sampling time, but efficiency falls because of process-management and communication overhead.